

Physics 155 Final Project Report Investigating Crazy Bounces

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1 Introduction

When a tennis ball is placed directly above a basketball and both are dropped from the same height, the tennis ball rebounds to a surprisingly high height, much higher than its original drop height. This project investigates the physics behind this “crazy bounce” using Newtonian mechanics, focusing on momentum and energy transfer during collisions.

2 Experimental Setup

For our setup, we used a tennis ball with a mass of 55g, a basketball with a mass of 545g, a 1-meter stick for distance calibration in Pasco Capstone, an iPhone camera for video recording, and the Pasco software for analysis. In each trial, we placed the tennis ball directly above the basketball and released both from rest. The basketball was held at a moderate height above the ground, high enough to minimize measurement uncertainty, but within the limits of our reach to ensure controlled release. We slightly varied the initial height across trials.

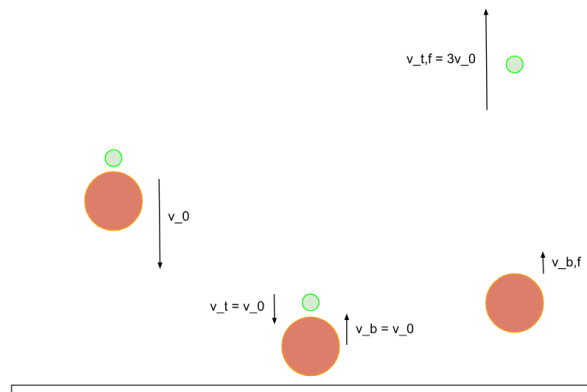


Figure 1: Experiment Setup

3 Theoretical Model

3.1 Model Assumptions and Implications

To model and analyze the two-ball drop system effectively, we made several simplifying assumptions to simplify computations and the derivation of the model.

Assumption	Justification and Impact
Negligible air resistance	Air resistance is minimal over short distances and moderate speeds. Ignoring it simplifies the model, but may slightly overestimate the rebound height due to ignored drag.
Perfectly elastic collisions	Assumes conservation of both kinetic energy and momentum. This simplifies analysis but overestimates rebound height since real collisions lose energy to heat and deformation.
1D vertical motion	Treating motion as purely vertical allows for a simple 1D model. Misalignment or lateral motion would introduce angular or horizontal effects not captured by the model.
No spin or rotation	Avoids dealing with angular momentum and torque. This simplifies modeling but excludes effects from spin that may slightly affect energy transfer.
Point masses	By treating balls as point masses, we avoid modeling internal deformation and rotational inertia. This simplifies derivation but ignores size-based energy loss.

3.2 Mathematical Derivation

We adopt the following sign conventions:

- **Downward direction** is taken as **positive**.
- **Upward direction** is taken as **negative**.

Let the variables be defined as follows:

m	mass of the tennis ball (0.055 kg)
M	mass of the basketball (0.545 kg)
g	acceleration due to gravity (9.8 m/s ²)
h_0	initial drop height
v	speed of both balls just before impact
v_m	velocity of the tennis ball before the collision
v_M	velocity of the basketball before the collision
v'_m	velocity of the tennis ball after the collision
h_r	rebound height of the tennis ball

1D Elastic Collision Formula

In a one-dimensional elastic collision, both momentum and kinetic energy are conserved. A well-known result of such collisions gives the final velocity of one object (object 1) after colliding with another (object 2):

$$v'_1 = \frac{(m_1 - m_2)}{m_1 + m_2} v_1 + \frac{2m_2}{m_1 + m_2} v_2$$

This formula is standard and derived from conservation laws; the proof is omitted here for brevity.

3.2.1 Application to the Two-Ball Drop Scenario

Just before impact with the ground, both balls are falling at speed $v = \sqrt{2gh_0}$. Using the convention that downward is positive:

- Basketball velocity after rebound: $v_M = -\sqrt{2gh_0}$
- Tennis ball velocity before collision: $v_m = +\sqrt{2gh_0}$

Applying the formula:

$$v'_m = \frac{(m - M)}{m + M} \cdot v_m + \frac{2M}{m + M} \cdot v_M = \sqrt{2gh_0} \cdot \left(\frac{-0.49 - 1.09}{0.6} \right) \approx -2.63\sqrt{2gh_0}$$

3.2.2 Rebound Height of the Tennis Ball

According to the law of conservation of mechanical energy, the kinetic energy of the tennis ball just after the collision is fully converted into gravitational potential energy at the peak of its bounce. Thus, we equate:

$$\frac{1}{2}mv_m'^2 = mgh_r$$

Canceling m from both sides:

$$\frac{1}{2}v_m'^2 = gh_r$$

Solving for h_r :

$$h_r = \frac{(v_m')^2}{2g}$$

Substituting $v'_m = -2.63\sqrt{2gh_0}$:

$$h_r = \frac{(-2.63\sqrt{2gh_0})^2}{2g} = \frac{6.92 \cdot 2gh_0}{2g} = 6.92h_0$$

Therefore, the rebound height of the tennis ball is approximately 6.92 times the original drop height. This dramatic rebound results from the efficient transfer of momentum and kinetic energy from the heavier basketball to the lighter tennis ball.

4 Experimental Verification with Capstone

4.1 Measurement Definitions

- **Initial height (h_0):** Distance from the release point to the ground.
- **Rebound height (h_r):** Maximum height reached by the tennis ball after the bounce.

4.2 Trial Data

Trial	Initial Height (h_0)	Rebound Height (h_r)	Ratio (h_r/h_0)
1	0.95 m	3.34 m	3.52
2	1.04 m	2.95 m	2.84
3	0.31 m	1.67 m	5.39
4	0.95 m	3.87 m	4.07
5	2.13 m	4.77 m	2.24
Average Rebound Factor			3.61

This average rebound factor of 3.61 is lower than the calculated rebound factor, revealing the limits of idealized assumptions.

4.2.1 Discussion

Real-world systems are rarely governed by ideal equations alone. Several factors contributed to the gap between theoretical and observed rebound heights:

- **Energy loss:** Collisions were not perfectly elastic. Some kinetic energy was dissipated as heat, sound, or internal deformation, especially at the contact points between the basketball and the floor and between the basketball and the tennis ball. Real-world materials have a coefficient of restitution less than 1, which leads to rebound velocities lower than the incident velocities, reducing the total post-collision energy available for vertical motion.
- **Finite mass ratio (10:1):** Although the basketball is about 10 times more massive than the tennis ball, this ratio is far from the infinite-mass limit assumed in simplified derivations. As a result, the basketball does not act as an immovable wall during the collision and instead recoils slightly, reducing the relative velocity and energy transferred to the tennis ball. Our theoretical prediction of $6.92h_0$, as opposed to $9h_0$, accounts for this more realistic mass ratio.
- **Point-mass approximation:** In our mathematical model, both balls are treated as point masses, meaning we ignore their finite size, deformation, and rotational inertia. This allows us to simplify the system to a one-dimensional motion governed by linear momentum and energy conservation. However, in reality, finite-size effects, such as where on the surfaces the balls make contact, can influence the collision outcome, especially if either ball rotates or shifts laterally.
- **No terminal velocity:** Our model assumes that both balls accelerate freely under gravity from rest without ever reaching terminal velocity. In practice, due to the relatively short fall distance (under 2.5 meters), the velocities involved are far below the threshold where air resistance balances gravitational force. Thus, this assumption holds well for the scope of our experiment, and neglecting terminal velocity introduces a negligible error.
- **Collision timing:** If the tennis ball contacts the basketball before it has fully rebounded from the ground, the relative velocity at impact is less than in the ideal case. Such mistiming can occur due to imperfect stacking or small initial gaps. This reduces the energy transfer efficiency, causing the tennis ball to rebound lower than predicted. This effect is particularly sensitive to experimental setup and alignment.
- **Rotational motion and drag:** Any slight horizontal misalignment at release may cause either ball, particularly the tennis ball, to spin or deviate from vertical motion. Rotational motion introduces angular momentum and increases the system's energy dissipation through air drag and surface interactions. Although air resistance is minor over small vertical distances, it contributes to small reductions in rebound height and makes trajectories harder to track precisely.
- **Measurement error:** Capstone video analysis, while highly effective for tracking motion, is constrained by factors such as frame rate, image resolution, motion blur, and manual calibration. Identifying exact moments of ground contact and maximum height involves frame-by-frame judgment, introducing uncertainty. Calibration with meter sticks may also cause scaling inaccuracies if the balls deviate from the camera's focal plane.

Despite these non-ideal conditions, the experiment confirms that significant energy transfer from the heavier ball to the lighter one occurs. An average rebound factor of 3.61 clearly reflects the underlying physics of the “super bounce.”

5 Limitations of Experimental Setup

While our analysis aimed to capture the core physical principles governing the super bounce, several limitations of the experimental apparatus and procedure may have influenced the results.

- **Stacking alignment:** Achieving perfect vertical alignment between the tennis ball and basketball is challenging. Even minor misalignments at release can lead to off-axis impacts, energy loss due to sideways motion, or rotational effects, all of which reduce the rebound height.

- **Consistent release conditions:** Releasing both balls simultaneously without initial spacing or disturbance requires precision. Manual releases often introduce small gaps or tilts that affect the sequence and timing of the collisions.
- **Limited drop height:** Our setup constrained the drop height to under 2.5 meters. A greater height could amplify the differences between ideal and real conditions (e.g., air drag or deformation), or provide clearer measurements, especially for high rebounds.
- **Calibration uncertainty:** The Capstone software relies on calibration using meter sticks or known references. If the ball moves slightly out of the calibrated plane during the bounce, the measurements may be distorted due to perspective or parallax error.
- **Frame resolution and frame rate:** Although adequate, the video frame rate may not always capture the exact moment of ground contact or peak rebound. This introduces timing inaccuracies and affects position extrapolation.
- **Environmental conditions:** Minor air currents, surface irregularities on the floor, or imperfections in the balls themselves (e.g., uneven wear or inflation) can alter bounce behavior, especially in a sensitive energy-transfer setup like this one.

These limitations, while not undermining the core findings, emphasize the importance of experimental control and repeatability when testing idealized physical models. With improved equipment—such as synchronized release mechanisms, motion sensors, or high-speed cameras—future iterations of the experiment could further reduce error margins.

6 Conclusion

By constructing a simplified theoretical model based on Newtonian mechanics and conservation laws, we predicted a rebound height of approximately $6.92h_0$. This result was derived under ideal assumptions: perfectly elastic collisions, infinite basketball mass, non-overlapping impacts, and strictly vertical motion. These conditions allowed us to isolate the key mechanism responsible for the so-called “super bounce”: the efficient transfer of momentum and energy from a heavy object to a much lighter one.

Our experimental trials, conducted using Capstone video analysis, showed rebound factors ranging between approximately 2.2 and 5.4. Although these values were lower than the theoretical maximum, they are consistent with a system in which many real-world complications play significant roles, such as energy dissipation, imperfect timing, finite mass ratios, and measurement limitations. Notably, the discrepancy between theoretical and observed results served not to invalidate the model but rather to highlight the sensitivity of the outcome to even minor deviations from ideal conditions.

This project illustrates a broader pedagogical value: Idealized models are not meant to mirror reality exactly, but to reveal and clarify the governing principles behind complex physical systems. In our case, the essential insight was upheld, that a small, light ball can rebound significantly higher after a collision with a heavy, rebounding object. The model provided a reference point against which the influence of non-ideal effects could be evaluated.

Furthermore, the process of refining our assumptions, quantifying our experimental errors, and critically analyzing the mismatch between prediction and observation helped deepen our understanding of the physical modeling itself. Rather than serving as a final answer, the model acted as a framework for investigation, one that is both explanatory and exploratory.

In conclusion, the super bounce is more than just a physics demonstration. It is a vivid example of how energy and momentum interact in multibody systems, and how even simple setups can yield rich and insightful dynamics when approached through careful analysis and critical experimentation.